



2016
FORM V
ANNUAL EXAMINATION

Monday 29th August, 12:55 p.m.

Chemistry

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using black pen
- Draw diagrams using pencil
- A Data Sheet and Periodic Table are provided at the back of this paper
- **Write your name and Master's initials on the multiple choice answer sheet, and at the top of Question 15, 17, 21 and 25.**
- **Remove the central staple before handing in paper**

Total marks (86)

This paper consists of two parts, **Part A** and **Part B**.

Part A

Total marks (12)

- Attempt ALL Questions
- Allow about 15 minutes for this Part.

Part B

Total marks (74)

- Attempt ALL questions
- Allow about 1 hour and 45 minutes for this Part.

CHECKLIST

Each boy should have the following:

1 Question Paper

1 Multiple Choice Answer Sheet

Chemistry Classes

5CY201 – EJS/LL	5CY202 - TW	5CY203 - AKBB	5CY204 - DGB
5CY205 - MRB	5CY206 – TW/LL	5CY207 - CRMR	5CY208 - ZI

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Part A

Total marks (12)

Attempt ALL Questions

Allow about 15 minutes for this Part

Use the multiple-choice Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill the response circle completely.

Sample $2 + 4 =$

(A) 2

(B) 6

(C) 8

(D) 9

☐ (A)

☒ (B)

☐ (C)

☐ (D)

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

☒ (A)

☒ (B)

☐ (C)

☐ (D)

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

☒ (A)

☒ (B) *correct*

☐ (C)

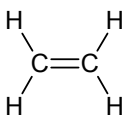
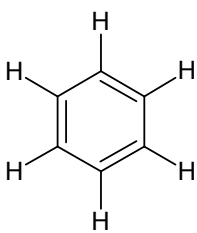
☐ (D)

- 1 Which of the following best defines the term “allotrope”?
- (A) Different structural forms of an element.
 - (B) Atoms with the same number of protons but differing number of neutrons.
 - (C) Compounds that have the same molecular formula but different structural formula.
 - (D) The different phases of a substance (e.g. gas, liquid or solid).
- 2 Which one of the following statements most accurately relates the properties of a liquid at room temperature to its vapour pressure?
- (A) A liquid with a high vapour pressure will probably have strong intermolecular forces and a low boiling point.
 - (B) A liquid with a high vapour pressure will probably have low surface tension and a high boiling point.
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- 3 Which one of the following processes is endothermic?
- (A) Combustion of natural gas.
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- 4 Which one of the following statements about the behaviour of a catalyst is correct?
- (A) A catalyst reacts with the products to speed up the reaction.
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5 Solder is an alloy of lead and tin. A useful property of solder is that:

- (A) it is lustrous, like lead and tin.
- (B) it conducts electricity, like lead and tin.
- (C) it has a high melting point.
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6 Which of the following compounds have the same empirical formula?

(i) 	(ii) $\text{H}-\text{C}\equiv\text{C}-\text{CH}_3$
(iii) 	(iv) $\text{H}-\text{C}\equiv\text{C}-\text{H}$

- (A) (i) and (iv)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) None of the above

7 Which of the following contains only molecules with dipole-dipole interactions as their main intermolecular force?

- (A) Hydrogen chloride, hydrogen sulfide, water.
- (B) Ammonia, phosphorus trifluoride, carbon dioxide.
- (C) Ammonia, sodium chloride, methane.
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8 The density of water at 4°C is 1.00 g.mL⁻¹. The density of ice at 0 °C is 0.917 g mL⁻¹. What volume change would occur if an ice cube of mass 7.40 g, initially at 0 °C, melted to form liquid water with a final temperature of 4 °C?

- (A) The volume would increase by 0.61 mL.
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- 9 Within a Group of the Periodic Table, which is the correct relationship between a property of a metal and the first ionization energy of the metal?
- (A) The larger the atom, the smaller the ionisation energy.
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 - (C) The smaller the amount of energy required to remove an electron from the metal, the greater the ionisation energy.
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- (A) The energy of an electron is variable, irrespective of its location.
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- 11 Identify which row below contains the correct information about the boiling and electrolysis of water.

	Type of separation	Amount of energy required	Result of Process
(A)	Boiling	Relatively large amount of energy	New chemical substance formed
(B)	Boiling	Relatively small amount of energy	No new chemical substance formed
(C)	Electrolysis	Relatively small amount of energy	New chemical substance formed
(D)	Electrolysis	Relatively large amount of energy	No new chemical substance formed

- 12 Which of the following statements about empirical formulae is correct?
- (A) They express the simplest ratio of protons to neutrons present in a compound.
 - (B) They provide information about the elements present in a compound but don't provide information about the number of particles.
 - (C) They can only be applied to ionic compounds.
 - (D) They show the number of atoms of different elements in a molecule.

Part B**Total marks (74)****Attempt ALL Questions**

Master's initials

Name

Answer the questions in the spaces provided.

Show **all** relevant working in questions involving calculations.**Question 13** (6 marks)**Marks**

(a) Write balanced chemical equations for the following reactions:

i. Decomposition of silver chloride.

2

ii. Reaction of aqueous solutions of barium nitrate and copper(II) sulfate.

2

(b) Write a balanced net ionic equation for the following reaction:

i. Reaction of aqueous solutions of lead(II) nitrate and potassium iodide.

2

Question 14 (5 marks)

Marks

During the course of this year you have performed a first-hand investigation to separate a naturally occurring mixture of sand, salt and water.

- (a) Identify the separation technique you used to perform the separation of the sand from salt water and state the physical property which enabled this technique.

2

- (b) One of your classmates was given a sample of this mixture and asked to perform a gravimetric analysis of it. He found that at 22 °C the sample contained 0.573 L of water (density at 22 °C is 0.9978 g mL⁻¹), 7.8 g of sand and 2.6 g of salt.

Determine the percentage composition by mass of **water** in this sample.

3

Question 15 (3 marks)

Marks

Consider the representation of an atom shown below.



Using the letters in the representation above, identify:

- (a) The number of protons in this atom.

1

- (b) The number of neutrons in this atom.

1

- (c) An expression for the number of electrons found in an ion of this element if the element belongs to group 5 of the periodic table.

1

Question 16 (3 marks)**Marks**

Identify the property or substance represented by the letters X, Y and Z in the table below.

Group	Metals	Metalloids	Non-metals
Appearance	lustrous	Low-sheen	dull
Electrical Conductivity	High	low	Y
Thermal Conductivity	high	high	low
Malleability	X	moderate	low
Density	high	intermediate	low
Boiling point	high	high	low
Tensile strength	High	variable	low
Examples	Zinc	Z	Argon

X : _____ **1**

Y : _____ **1**

Z : _____ **1**

Master's initials

Name

Question 17 (5 marks)**Marks**

Both ionic compounds and metallic elements are most commonly found as lattice structures. Compare these two types of lattice structures in terms of their structure and bonding.

5**Question 18** (2 marks)

Chemical reactions involve energy transformations. Describe a chemical reaction that you have observed that involves the transformation of light energy to chemical potential energy and provide an equation for this process.

2

Question 19 (6 marks)**Marks**

Hydrogen fluoride can be prepared according to the equation:



It is estimated that the activation energy for this reaction to proceed is 160 kJ mol^{-1} .

- (a) Draw a labelled energy profile diagram to represent this reaction.

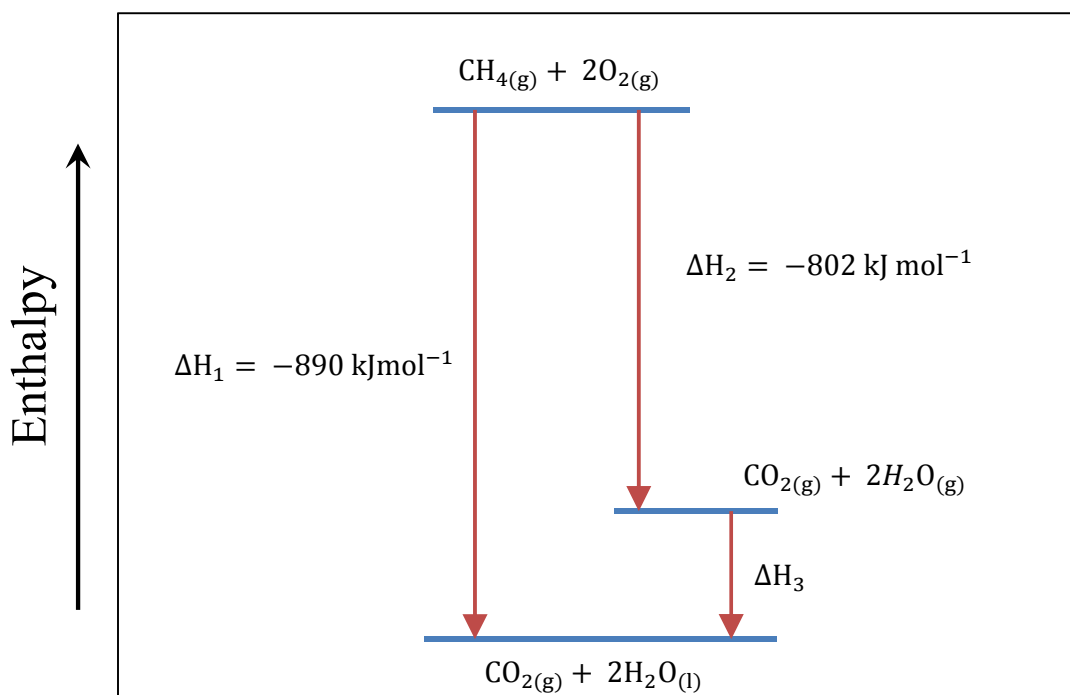
3

- (b) Using collision theory, explain what the effect on the reaction rate would be if the temperature was increased.

3

Question 20 (7 marks)**Marks**

Hess's Law states that the total enthalpy change during the complete course of a chemical reaction is the same irrespective of the pathway taken i.e. whether the reaction is made in one step or in several steps. Consider the following example for the complete combustion of methane.



- (a) If Hess's Law holds for the above example, calculate ΔH_3 .

2

Question 20 continued on next page.

Question 20 continued.

Marks

- (b) Identify the fourth member of the homologous series of which methane is the first member.

1

- (c) Identify the original source of the chemical potential energy found in methane.

1

- (d) Write an equation for the incomplete combustion of methane.

1

- (e) Explain why the incomplete combustion of hydrocarbons like methane is undesirable.

2

Master's initials

Name

Question 21 (3 marks)**Marks**

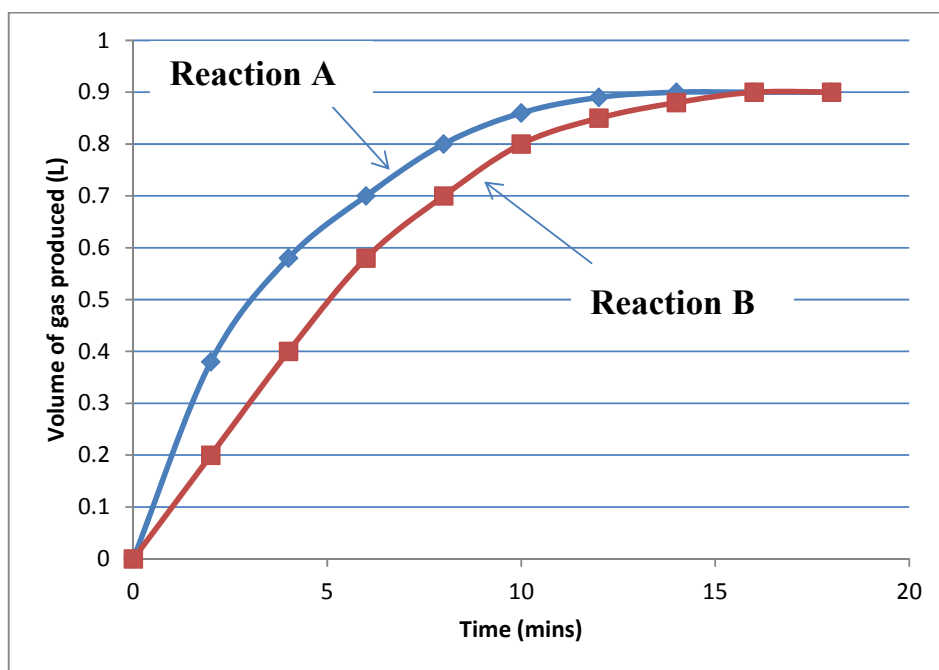
With reference to the tabulated first ionisation energies below, explain the differences in reactivity between potassium and rubidium.

Metal	First Ionisation Energy (kJ mol ⁻¹)
Potassium	425
Rubidium	403

3

Question 22 (6 marks)**Marks**

Excess copper(II) carbonate was placed in a conical flask and 100.0 mL of dilute sulfuric acid was added. A reaction occurred and the gas produced was collected. The volume of gas collected at 100 kPa and 25 °C is shown in the graph below for Reaction A.



- (a) Calculate the concentration of the sulfuric acid.

3

Question 22 continued on next page.

Question 22 continued.

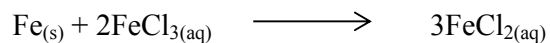
Marks

- (b) A second reaction with the same mass of copper(II) carbonate and volume of acid was carried out and the results shown in Reaction B. Identify the possible conditions that could have changed to produce this graph and explain their effects.

3

Question 23 (7 marks)

Solutions of $\text{FeCl}_{2(\text{aq})}$ oxidise easily in the presence of oxygen to produce $\text{FeCl}_{3(\text{aq})}$. Iron filings can be added to stabilise these solutions by reducing any $\text{FeCl}_{3(\text{aq})}$ formed back to $\text{FeCl}_{2(\text{aq})}$ according to the following equation.



- (a) Write the net ionic equation for this reaction.

1

- (b) Write the reduction and oxidation half equations for these reactions.

- (i) Reduction

1

- (ii) Oxidation

1

- (c) Calculate the mass of FeCl_3 can be reduced completely to FeCl_2 by 1.00 g of $\text{Fe}_{(\text{s})}$.

4

Question 24 (4 marks)**Marks**

2.67 L of vapourised phosphorus molecules are reacted with 8.01 L of oxygen gas at a constant temperature and pressure to produce 2.67 L of a gaseous oxide of phosphorus having a molar mass of 220 g mol^{-1} .

- (a) Balance the given equation below for the reaction.



- (b) Determine the formula of the oxide of phosphorus. Show working.

2

- (c) Determine the molecular formula of the vapourised phosphorus molecules.

1

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Master's initials

Name

Question 25 (6 marks)**Marks**

(a) Construct Lewis dot diagrams of the following molecules

(i) methane

1

(ii) water

1(iii) hydrogen sulfide (H_2S)**1**

(b) Explain why the boiling points of water and hydrogen sulfide differ.

3

Question 26 (5 marks)

Marks

In an experiment, a boy added 8.00 g of ammonium nitrate to 100 g of water.

- (a) If the $\Delta_{\text{sol}}H$ for ammonium nitrate is + 25.41 kJ mol⁻¹ and the water was initially 25.0 °C, calculate the final temperature of the resulting solution.

3

- (b) The boy actually measured a final temperature of 22.1 °C. Suggest two possible reasons as to why there is a difference between these results.

2

Question 27 (6 marks)

Marks

Compound **A** is a white insoluble mineral made of four elements. When 10.000 g of compound **A** is reacted with 2 M HCl, 2.6886 L of CO₂ (measured at 25 °C and 100 kPa), is evolved. Na₂SO₄ is then added to the solution forming 7.383 g of compound **B** as a precipitate; compound **C** remains in solution.

The solution of compound **C** is neutralised with 2 M NaOH forming 3.1628 g of a precipitate of compound **D**, which was filtered. The 3.1628 g of Compound **D** was then heated to produce 2.1859 g of compound **E** and water. Compound **E** was heated at 2300 °C with excess carbon to give CO and 1.3183 g of pure metal **F**.

What are the formulae of compounds **A-F**? Show working.

6

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Chemistry

Data Sheet

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0 °C (273 K)	22.71 L
at 25 °C (298 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \quad q = mC\Delta T$$

Standard Potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}_{(\text{s})}$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}_{(\text{s})}$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}_{(\text{s})}$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}_{(\text{s})}$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}_{(\text{s})}$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}_{(\text{s})}$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}_{(\text{s})}$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})} + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}_{(\text{s})}$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}_{(\text{s})}$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}_{(\text{s})}$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}_{(\text{s})}$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}_{(\text{s})}$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})}$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_{2(\text{g})} + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.34 V
$\frac{1}{2} \text{O}_{2(\text{g})} + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.52 V
$\frac{1}{2} \text{I}_{2(\text{s})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2} \text{I}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}_{(\text{s})}$	0.80 V
$\frac{1}{2} \text{Br}_{2(\text{l})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2} \text{Br}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2} \text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2} \text{H}_2\text{O}$	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2} \text{F}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS																					
1 H 1.008 Hydrogen	KEY																				
3 Li 6.941 Lithium	Atomic Number Symbol Standard Atomic Weight Name																				
	4 Be 9.012 Beryllium	79 Au 197.0 Gold																			
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium					21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc						
19 K 39.10 Potassium	20 Ca 40.08 Calcium	38 Sr 87.61 Strontium	37 Rb 85.47 Rubidium	55 Cs 132.9 Caesium	87 Fr 223.0 Francium	89-103 Actinoids	104 Rf 261.1 Rutherfordium	105 Db 262.1 Dubnium	106 Sg 266.1 Seaborgium	107 Bh 264.1 Bohrium	108 Hs 277.1 Hassium	109 Mt 268.1 Meitnerium	110 Ds 271.1 Darmstadtium	111 Rg 272.1 Roentgenium	112 Cn 285.1 Copernicium						
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57-71 Lanthanoids				71 Lu 174.9 Lutetium	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po 209.0 Polonium	85 At 210.0 Astatine	86 Rn 222.0 Radon
87 Fr 223.0 Francium	88 Ra 226.0 Radium	89-103 Actinoids				103 La 138.9 Lanthanum	104 Ce 140.1 Cerium	105 Pr 140.9 Praseodymium	106 Nd 144.2 Neodymium	107 Pm 144.9 Promethium	108 Sm 150.4 Samarium	109 Eu 151.9 Europium	110 Gd 157.3 Gadolinium	111 Tb 158.9 Terbium	112 Dy 162.5 Dysprosium	113 Ho 164.9 Holmium	114 Er 167.3 Erbium	115 Tm 168.9 Thulium	116 Yb 173.0 Ytterbium	117 Lu 174.9 Lutetium	118 Og 289.1 Oganesson

57	La	138.9	Lanthanum
58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium
61	Pm		Promethium
62	Sm	150.4	Samarium
63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium
68	Er	167.3	Erbium
69	Tm	168.9	Thulium
70	Yb	173.1	Ytterbium
71	Lu	175.0	Lutetium

89	Ac	Actinium
90	Th	232.0 Thorium
91	Pa	231.0 Protactinium
92	U	238.0 Uranium
93	Np	Neptunium
94	Pu	Plutonium
95	Am	Americium
96	Cm	Curium
97	Bk	Berkelium
98	Cf	Californium
99	Es	Einsteinium
100	Fm	Fermium
101	Md	Mendelevium
102	No	Nobelium
103	Lr	Lawrencium

Standard atomic weights are abridged to four significant figures.

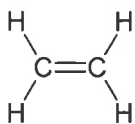
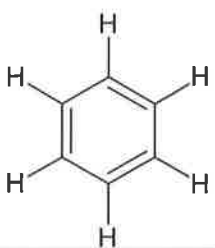
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.

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 - (B) The greater the reactivity of the metal, the greater the ionisation energy.
 - (C) The smaller the amount of energy required to remove an electron from the metal, the greater the ionisation energy.
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	Type of separation	Amount of energy required	Result of Process
(A)	Boiling	Relatively large amount of energy	New chemical substance formed
(B)	Boiling	Relatively small amount of energy	No new chemical substance formed
(C)	Electrolysis	Relatively small amount of energy	New chemical substance formed
(D)	Electrolysis	Relatively large amount of energy	No new chemical substance formed

- 12 Which of the following statements about empirical formulae is correct?
- (A) They express the simplest ratio of protons to neutrons present in a compound.
 - (B) They provide information about the elements present in a compound but don't provide information about the number of particles.
 - (C) They can only be applied to ionic compounds.
 - (D) They show the number of atoms of different elements in a molecule.

Part B**Total marks (74)****Attempt ALL Questions**

ALBBS

 Master's initials

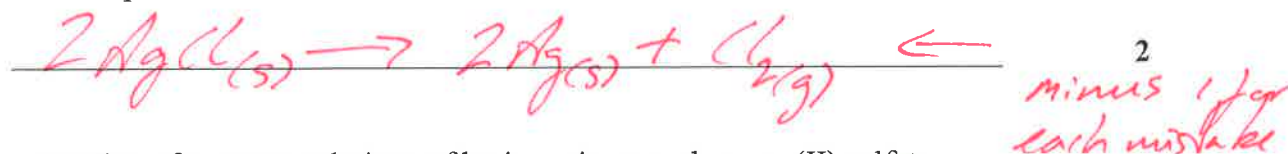
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Answer the questions in the spaces provided.

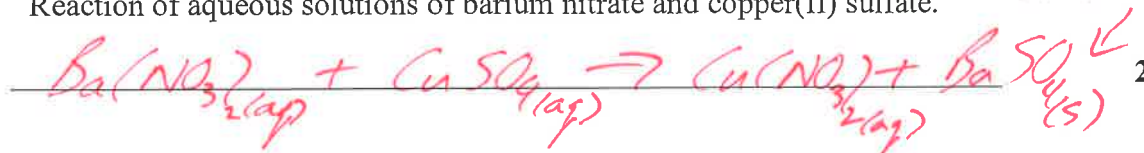
Show **all** relevant working in questions involving calculations.**Question 13 (6 marks)****Marks**

(a) Write balanced chemical equations for the following reactions:

i. Decomposition of silver chloride.

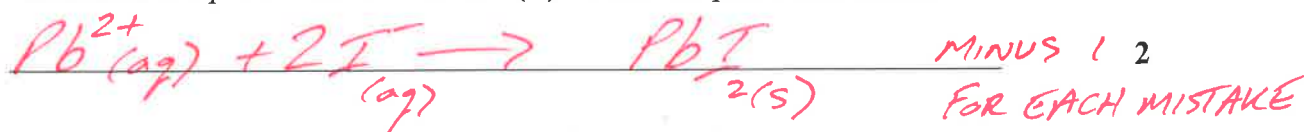


ii. Reaction of aqueous solutions of barium nitrate and copper(II) sulfate.



(b) Write a balanced net ionic equation for the following reaction:

i. Reaction of aqueous solutions of lead(II) nitrate and potassium iodide.



Many boys had several 'bites' at the cherry, writing the equation in different forms. Boys were penalised for each mistake they made, although not for writing several equations. This will not necessarily be the case in future exams (i.e. HSC).

Question 14 (5 marks)**Marks**

During the course of this year you have performed a first-hand investigation to separate a naturally occurring mixture of sand, salt and water.

- (a) Identify the separation technique you used to perform the separation of the sand from salt water and state the physical property which enabled this technique.

Filtration 1 MARK 2

Particle Size/Solubility 1 MARK

- (b) One of your classmates was given a sample of this mixture and asked to perform a gravimetric analysis of it. He found that at 22 °C the sample contained 0.573 L of water (density at 22 °C is 0.9978 g mL⁻¹), 7.8 g of sand and 2.6 g of salt.

Determine the percentage composition by mass of **water** in this sample.

Mass of water = 0.9978 × 0.573 × 10³
= 5.7174 × 10² g 1 MARK 3

Total mass = 5.7174 × 10² + 7.8 + 2.6 1 MARK
= 582.14 g

% Comp = $\frac{571.74}{582.14} \times 100 = 98.2\%$ 1 MARK

Question 15 (3 marks)

Marks

Consider the representation of an atom shown below.



Using the letters in the representation above, identify:

- (a) The number of protons in this atom.

 Z 1

- (b) The number of neutrons in this atom.

 $A - Z$ 1

- (c) An expression for the number of ~~valence~~ electrons found in an ion of this element if the element belongs to group 5 of the periodic table.

 $Z + 3$ 1

valence electrons 8, 18, 32, ...

Question 16 (3 marks)

Marks

Identify the property or substance represented by the letters X, Y and Z in the table below.

Group	Metals	Metalloids	Non-metals
Appearance	lustrous	Low-sheen	dull
Electrical Conductivity	High	low	Y
Thermal Conductivity	high	high	low
Malleability	X	moderate	low
Density	high	intermediate	low
Boiling point	high	high	low
Tensile strength	High	variable	low
Examples	Zinc	Z	Argon

X: High/good 1

Y: Any Low/poor 1

Z: Low/poor any 1

Question 17 (5 marks)**Marks**

Both ionic compounds and metallic elements are most commonly found as lattice structures. Compare these two types of lattice structures in terms of their structure and bonding.

Marked holistically.

5 marks required detailed discussion of S&B of both metals and ionic compounds, plus gives a direct comparison between them. Following codes for used to indicate what was missing or insufficiently covered (-1 per item):

C - explicit comparison missing

IB – ionic bonding

IS – ionic structure

MB – metallic bonding

MS – metallic structure

D – lacks depth

Sample answer

Metallic elements' structure involves a 3-D lattice of metal cations in a sea of delocalised electrons. The bonding is due to the electrostatic attraction between the metal's positive nuclei and the electrons. Ionic compounds also have a 3D lattice structure – but they have an array of repeating units of alternating cations and anions. These ions are also bonded by electrostatic attraction but this time it is between the fixed cations and anions and there are no free electrons. This ionic bonding makes the lattice structure very rigid (shatters when hit) compared to the easily malleable lattice structure of metals (as the free electrons can readily relocate to ensure the metallic bonds stay in place as the metal is beaten).

Question 18 (2 marks)

Chemical reactions involve energy transformations. Describe a chemical reaction that you have observed that involves the transformation of light energy to chemical potential energy and provide an equation for this process.

1 mark – description of reaction (e.g. decomposition of silver halides; photosynthesis)

1 mark – balanced chemical equation (note: word equations not accepted)

Question 19 (6 marks)**Marks**

Hydrogen fluoride can be prepared according to the equation:



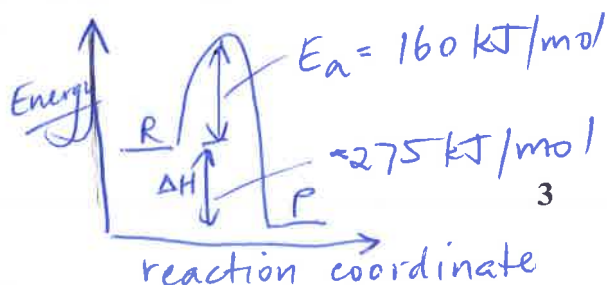
It is estimated that the activation energy for this reaction to proceed is 160 kJ mol^{-1} .

- (a) Draw a labelled energy profile diagram to represent this reaction.

1 mark – exothermic diagram

1 mark – labelling $\Delta H = -275$ on diagram

1 mark – labelling activation energy



- (b) Using collision theory, explain what the effect on the reaction rate would be if the temperature was increased.

3

1 mark – more frequent collisions

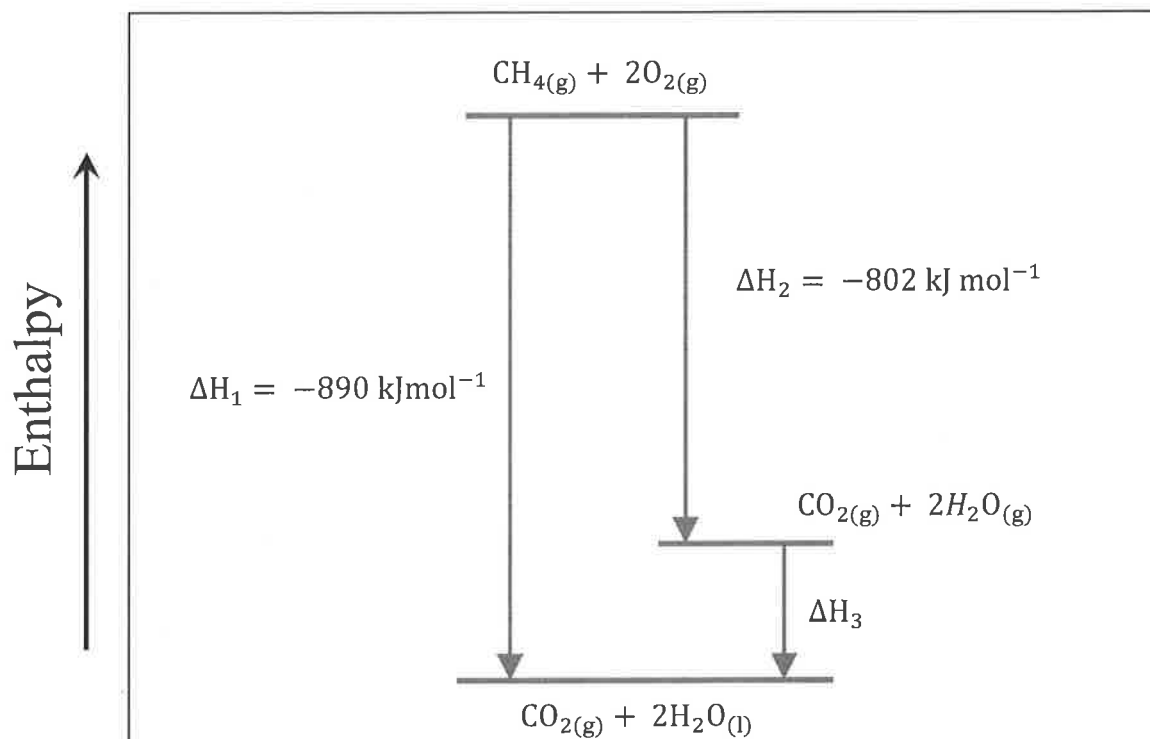
1 mark – more particles have sufficient energy to overcome activation barrier

1 mark – increase in reaction rate

Question 20 (7 marks)

Marks

Hess's Law states that the total enthalpy change during the complete course of a chemical reaction is the same irrespective of the pathway taken i.e. whether the reaction is made in one step or in several steps. Consider the following example for the complete combustion of methane.



- (a) If Hess's Law holds for the above example, calculate ΔH_3 .

2 marks - calculating -88 kJ/mol

1 mark for +88 kJ/mol

2

- (b) Identify the fourth member of the homologous series of which methane is the first member.

butane

1

- (c) Identify the original source of the chemical potential energy found in methane.

The sun

1

- (d) Write an equation for the incomplete combustion of methane.

Any correctly balanced chemical equation showing production of either C or CO

1

e.g. $\text{CH}_4 + 3/2 \text{O}_2 \rightarrow \text{CO} + 2 \text{H}_2\text{O}$ – states were ignored.

- (e) Explain why the incomplete combustion of hydrocarbons like methane is undesirable.

2 marks – explanation of two problems with incomplete combustion e.g. soot or CO or inefficient energy output. Needed to have detail of why problem exists e.g. needed more than CO is toxic; needed to explain how it caused oxygen shortage due to binding to haemoglobin.

1 mark – one of the above well explained or less detail on 2 problems.

2

Master's initialsName**Question 21** (3 marks)**Marks**

With reference to the tabulated first ionisation energies below, explain the differences in reactivity between potassium and rubidium.

Metal	First Ionisation Energy (kJ mol ⁻¹)
Potassium	425
Rubidium	403

- Explain first ionization energy
reduced

- Correlate first ionization energy
to increased reactivity

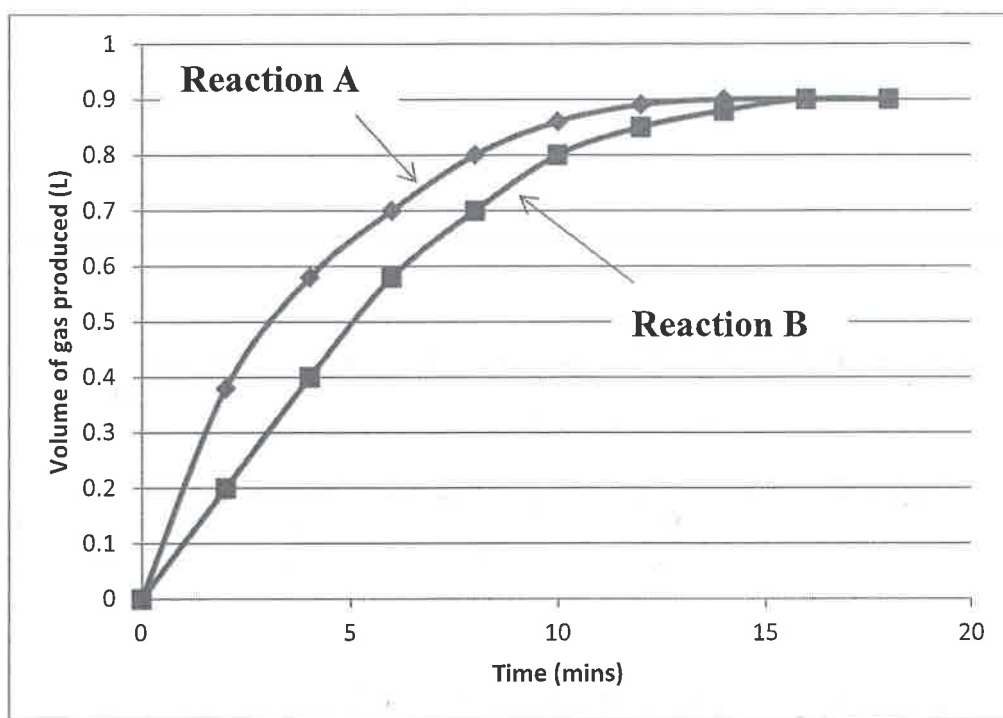
- refer to value with
rubidium having lower ionization
energy and hence higher reactivity.

3

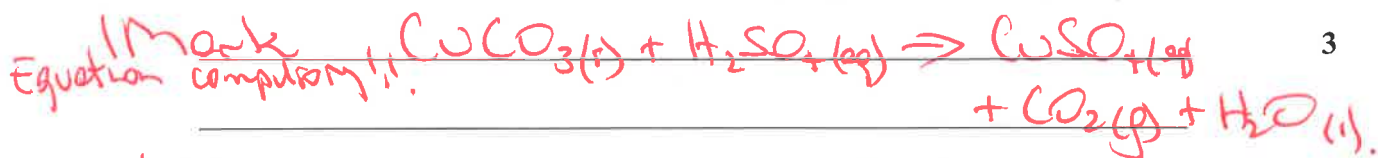
Question 22 (6 marks)

Marks

Excess copper(II) carbonate was placed in a conical flask and 100.0 mL of dilute sulfuric acid was added. A reaction occurred and the gas produced was collected. The volume of gas collected at 100 kPa and 25 °C is shown in the graph below for Reaction A.



(a) Calculate the concentration of the sulfuric acid.



1 Mark. $n(\text{CO}_2) = \frac{0.90}{24.79} = 0.0360 \text{ mol}$

1 Mark $n_{\text{H}_2\text{SO}_4} = n_{\text{CO}_2} = 0.0360 \text{ mol}$

$[\text{H}_2\text{SO}_4] = \frac{0.03630}{0.100} = 0.3630 \text{ M}$

Correct units required!

Question 22 continued on next page.

- (b) A second reaction with the same mass of copper(II) carbonate and volume of acid was carried out and the results shown in Reaction B. Identify the possible conditions that could have changed to produce this graph and explain their effects.

2 Reasons 1 mark each

1 Explanation 1 mark

3

Marks lost for incorrect reasons or explanations.

Reasons

- ~~Increased~~ Decreased temp of solution
- Decreased surface area of copper (II) carbonate

~~Decreased~~

Not Decreased concentration as the volume added was fixed (see comment above) and this would have led to a smaller volume of gas produced.

Explanation mentions - frequency of collision
- energy per collision.

Question 23 (7 marks)

Solutions of $\text{FeCl}_{2(\text{aq})}$ oxidise easily in the presence of oxygen to produce $\text{FeCl}_{3(\text{aq})}$. Iron filings can be added to stabilise these solutions by reducing any $\text{FeCl}_{3(\text{aq})}$ formed back to $\text{FeCl}_{2(\text{aq})}$ according to the following equation.



- (a) Write the net ionic equation for this reaction.



1

- (b) Write the reduction and oxidation half equations for these reactions.

- (i) Reduction



1

- (ii) Oxidation



1

- (c) Calculate the mass of FeCl_3 can be reduced completely to FeCl_2 by 1.00 g of $\text{Fe}_{(\text{s})}$.

$$n_{\text{Fe}} = \frac{1.00}{55.85} = 0.01791 \text{ mol}$$

4

$$n_{\text{FeCl}_3} = 2 \times n_{\text{Fe}} = \frac{2.00}{55.85} = 0.03581 \text{ mol}$$

$$m_{\text{FeCl}_3} = n \times M = \frac{2.00}{55.85} \times 162.2$$

$$= 4.57 \text{ g} \rightarrow 5.8 \text{ g}$$

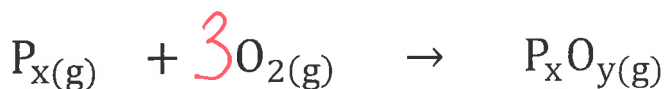
$$\begin{array}{r} 220 \\ 96 \\ \hline 124 \end{array}$$

Question 24 (4 marks)

Marks

2.67 L of vapourised phosphorus molecules are reacted with 8.01 L of oxygen gas at a constant temperature and pressure to produce 2.67 L of a gaseous oxide of phosphorus having a molar mass of 220 g mol^{-1} .

- (a) Balance the given equation below for the reaction.



1

- (b) Determine the formula of the oxide of phosphorus. Show working.

1 mark $y = \text{number of oxygens} = 2 \times 3 = 6$

$\times$ mass of 6 oxygens = $6 \times 16 = 96$

$\therefore$ mass of phosphorus = $220 - 96 = 124$

$\times$ number of phosphorus = $\frac{124}{31} = 4$

1 mark $\times$ Formula = P_4O_6

2

- (c) Determine the molecular formula of the vapourised phosphorus molecules.



1

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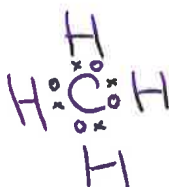
Master's initialsName

Marks

Question 25 (6 marks)

(a) Construct Lewis dot diagrams of the following molecules

(i) methane



1

(ii) water



1

(iii) hydrogen sulfide (H₂S)

1

(b) Explain why the boiling points of water and hydrogen sulfide differ.

① Water has hydrogen bonding

3

② H₂S has dipole-dipole & dispersion
~~mainly~~ had to say dipole dipole

③ more energy to disrupt water's intermolecular attraction therefore higher melting point.

Question 26 (5 marks)

Marks

In an experiment, a boy added 8.00 g of ammonium nitrate to 100 g of water.

- (a) If the $\Delta_{\text{sol}}H$ for ammonium nitrate is + 25.41 kJ mol⁻¹ and the water was initially 25.0 °C, calculate the final temperature of the resulting solution.

$$\begin{aligned}
 n_{\text{NH}_4\text{NO}_3} &= \frac{8.0}{80.052} \quad (1) \\
 &= 0.099935 \\
 q &= mc\Delta T \\
 -2539.348 &= 100 \times 4.18 \times \Delta T \\
 \Delta T &= \frac{-2539.348}{4.18} \\
 &= -6.074996 \\
 \Delta H_{\text{sol}} &= \frac{-q}{n} \quad (1) \\
 254100 &= \frac{-q}{n} \\
 q &= -2539.348 \text{ J} \\
 T_F - T_i &= \Delta T \\
 T_F &= 25 - 6.074996 \\
 &= 18.9^\circ\text{C} \quad (1)
 \end{aligned}$$

- (b) The boy actually measured a final temperature of 22.1 °C. Suggest two possible reasons as to why there is a difference between these results.

① temperature was gained from the surroundings

② impurities in ammonium nitrate

NOTE: for a) many used 4.18×10^3 but still left ΔH in kJ which gave the wrong value.

b) poor equipment and misreadings not accepted
Have to assume the experiment was conducted properly and carefully.

Question 27 (6 marks)

Marks

Compound A is a white insoluble mineral made of four elements. When 10.000 g of compound A is reacted with 2 M HCl, 2.6886 L of CO_2 (measured at 25 °C and 100 kPa), is evolved. Na_2SO_4 is then added to the solution forming 7.383 g of compound B as a precipitate; compound C remains in solution.

The solution of compound C is neutralised with 2 M NaOH forming 3.1628 g of a precipitate of compound D, which was filtered. The 3.1628 g of Compound D was then heated to produce 2.1859 g of compound E and water. Compound E was heated at 2300 °C with excess carbon to give CO and 1.3183 g of pure metal F.

What are the formulae of compounds A-F? Show working.

① A must contain a carbonate as CO_2 produced. 6

$$n_{\text{CO}_2} = 0.10845 \text{ moles.}$$

A = carbonate

B = sulfate

C = sulfate or chloride.

→ Then work bottom up.

$$\text{mass of O}_2 \text{ lost} = 2.1859 - 1.3183$$

D = hydroxide

$$= 0.8676$$

E = oxide

$$\therefore n_{\text{O}} = 0.054225$$

F = pure metal.

so if 1:1 oxygen: metal then 0.054225 moles of metal.

$$\therefore \text{Molar mass metal} = \frac{\text{mass pure metal}}{\text{moles of O}} = \frac{1.3183}{0.054225} = 24.31$$

$\therefore \text{Mg}$ (1)

so F = Mg

ratio Mg: CO_3

E = MgO (1)

1:2

D = Mg(OH)_2

$\therefore \text{A} = \text{Mg}(\text{CO}_3)_2 \times$

B = $\times \text{SO}_4$

assuming X has a minus 2 charge.

C = MgSO_4 or MgCl_2 (1)

$$\text{so moles } X = 0.054225$$

$$\therefore \text{M}_B = \frac{7.383}{0.054225}$$

$$= 136.149, \text{ CRMR}$$

$$\text{MwSO}_4 = 96.07 \therefore \text{M}_X = 40.079 = \text{Ca}$$

(1)

B = CaSO_4

(1) A = $\text{MgCa}(\text{CO}_3)_2$

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